

Investigation on the Transmission Coefficient of Linear Wave Propagating Over Submerged Barrier

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ABSTRACT

With the global mean sea level rising, coastal regions will be flooded by causing substantial domestic gross losses. In tropical and sub-tropical regions, coral reefs exist along coastlines around the world. Most of the energy in the waves would be eliminated by coming across the coastal barrier. The extended lagoons and steep flanks of most fringing reefs produce unique surf zone processes and they are the natural ports for the fishermen and the boats with small draft.

Using this analytical solution, the global characteristics of the wave propagating over submerged plate barrier considering the geometric and the incident wave parameters are analyzed with great efforts. It shows that the width of the plate barrier and the depth of the shallow water play important roles to eliminate the incident wave. The wave parameters are changed remarkably over the submerged plate barrier besides the wave period or frequency. The wave transmission coefficients are also obtained by contrasting the amplitudes of the incident wave and the diffracted one without breaking.

However, the wave breaking plays more significant role to eliminate the incident wave than the submerged barrier. The role of the submerged barrier is mostly under-estimated for reflecting and dissipating incident wave because the wave breaking and the turbulent would dissipate the wave energy. The wave breaking and the turbulent which are not included in this potential fluid model and they should be investigated in detail in the future.

KEY WORDS: Submerged rectangle barrier; Breaking wave; Eigenfunction expansion; Global analysis.

INTRODUCTION

With the global mean sea level rising, coastal regions will be flooded by causing substantial domestic gross losses (Hinkel, et al. 2014). In tropical and sub-tropical regions, coral reefs exist along coastlines around the world (Roeber, et al., 2012). Most of the energy in the waves would be eliminated by coming across the coastal barrier (Ferrario, et al., 2014; Zhang, et al., 2012). The extended lagoons and steep flanks of most fringing reefs produce unique surf zone processes and they are the natural ports for the fishermen and the boats with small draft (Huang et al., 2015).

For the problems of the wave runup over the beach with a slope or over the steep offshore slope, some experimental results provide interesting information such as the relation between the incident wave and the runup, which can be applied to practical situations to have a first guess of the excepted hazard (Yamazaki, et al., 2011).

Compared with the physical experiments, numerical simulations and analytical solutions with various methods are powerful and popular tools for its lower expenses. Numerical methods including finite volume-finite difference hybrid model (Olabarrieta, et al., 2011), Boussinesq model (Yao, et al., 2012) and other models involving turbulent structures (Zhou, et al., 2014) are developed to simulate the wave propagating over beaches or reef lagoons. Analytical methods, including Schwinger's variational formulation and eigenfunction expansion matching method (Zheng, et al., 2007) and approximate formulae (Churaev, et al., 2015), are classical and convenient methods to analyze the interaction between the waves and the marine obstacles.

For global analysis, which means a wide range of parameters, it is very expensive to carry out physical experiments, and it is time consuming to employ the computational fluid dynamics tools to investigate the transmission coefficients. In this paper, the linear wave diffracted by submerged barrier with steep flanks, which can be simplified as vertical faces, is solved. The semi-analytical method and the tool of dimensional analyses are used here because its accuracy and efficiency are higher than numerical methods and physical experiments. The incident wave is linear and the method of separation of variables and the eigenfunction expansion matching (Zheng, et al., 2007; Koley, et al., 2015) is used to solve the diffracted wave for the existing of the submerged plate barrier with finite water depths and vertical flanks.

The main purpose of this paper is to analyze the non-dimensional parameters and investigate globally the wave transmission coefficient over submerged plate barrier with finite water.

MATHEMATICAL MODEL

If the length of obstacle in the direction vertical to the orientation waves propagating along is assumed to be infinite, the issue that waves running over submerged rectangle barrier can be simplified as two dimensional, which is shown in Fig. 1 The whole domain can be divided into three subdomains, i.e., region 1, region 2 and region 3. The water depth and the height of the barrier are h_1 and d_1 , respectively. The